

Binary Addition

It is a key for binary subtraction, multiplication, division. There are four rules of binary addition.

Case	A	+	B	Sum	Carry
1	0	+	0	0	0
2	0	+	1	1	0
3	1	+	0	1	0
4	1	+	1	0	1

In fourth case, a binary addition is creating a sum of (1 + 1 = 10) i.e. 0 is written in the given column and a carry of 1 over to the next column.

Example – Addition

$$\begin{array}{r} 0011010 + 0011100 = 00100110 \\ \begin{array}{r} 11 \text{ carry} \\ 0011010 = 26_{10} \\ + 0001100 = 12_{10} \\ \hline 0100110 = 38_{10} \end{array} \end{array}$$

Binary Subtraction

Subtraction and Borrow, these two words will be used very frequently for the binary subtraction. There are four rules of binary subtraction.

Case	A	-	B	Subtract	Borrow
1	0	-	0	0	0
2	1	-	0	1	0
3	1	-	1	0	0
4	0	-	1	1	1

Example – Subtraction

$$\begin{array}{r} 0011010 - 0011100 = 00001110 \\ \begin{array}{r} 11 \text{ borrow} \\ 00\cancel{1}1010 = 26_{10} \\ - 0001100 = 12_{10} \\ \hline 0001110 = 14_{10} \end{array} \end{array}$$

Binary Multiplication

Binary multiplication is similar to decimal multiplication. It is simpler than decimal multiplication because only 0s and 1s are involved. There are four rules of binary multiplication.

Case	A	x	B	Multiplication
1	0	x	0	0
2	0	x	1	0
3	1	x	0	0
4	1	x	1	1

Example – Multiplication

Example:

$$0011010 \times 001100 = 100111000$$

$$\begin{array}{r} 0011010 = 26_{10} \\ \times 0001100 = 12_{10} \\ \hline 0000000 \\ 0000000 \\ 0011010 \\ 0011010 \\ \hline 0100111000 = 312_{10} \end{array}$$

Binary Division

Binary division is similar to decimal division. It is called as the long division procedure.

There are four parts in any division: Dividend, Divisor, quotient, and remainder.

Input A	Input B	Divide (D) A/B
0	0	Not defined
0	1	0
1	0	Not defined
1	1	1

The result is always not defined, whenever the divisor is 0.

Example – Division

$$101010 / 000110 = 000111$$

$$\begin{array}{r}
 111 = 7_{10} \\
 000110 \overline{) 101010} = 42_{10} \\
 \underline{-110} = 6_{10} \\
 1001 \\
 \underline{-110} \\
 110 \\
 \underline{-110} \\
 0
 \end{array}$$

Practice:

1. 101101/ 101
2. 1111000/ 100
3. 101011/ 101

Practice

Addition

1. $111 + 111$
2. $11.101 + 101.100$
3. $11101 + 1010$
4. $11011 + 1011 + 1101$

Solution:

Subtraction:

1. $1001 - 11$
2. $1101 - 101$
3. $10001 - 100$
4. $11011 - 1011$
5. $1101.01 - 101.11$

Solution:

Subtraction using 1's complement

These are the following steps to subtract two binary numbers using 1's complement

- In the first step, find the 1's complement of the subtrahend.
- Next, add the complement number with the minuend.
- If got a carry, add the carry to its LSB. Else take 1's complement of the result which will be negative

Example 1: 10101 - 00111

We take 1's complement of subtrahend 00111, which comes out 11000. Now, sum them. So,

$$10101 + 11000 = 1\ 01101.$$

In the above result, we get the carry bit 1, so add this to the LSB of a given result, i.e., $01101 + 1 = 01110$, which is the answer.

Example 2: 10101 - 10111

We take 1's complement of subtrahend 10111, which comes out 01000. Now, add both of the numbers. So,

$$10101 + 01000 = 11101.$$

In the above result, we didn't get the carry bit. So calculate the 1's complement of the result, i.e., 00010, which is the negative number and the final answer.

Subtraction using 2's complement

These are the following steps to subtract two binary numbers using 2's complement

- In the first step, find the 2's complement of the subtrahend.
- Add the complement number with the minuend.
- If we get the carry by adding both the numbers, then we discard this carry and the result is positive else take 2's complement of the result which will be negative.

Example 1: 10101 - 00111

We take 2's complement of subtrahend 00111, which is 11001. Now, sum them. So,

$$10101 + 11001 = 1\ 01110.$$

In the above result, we get the carry bit 1. So we discard this carry bit and remaining is the final result and a positive number.

Example 2: 10101 - 10111

We take 2's complement of subtrahend 10111, which comes out 01001. Now, we add both of the numbers. So,

$$10101 + 01001 = 11110.$$

In the above result, we didn't get the carry bit. So calculate the 2's complement of the result, i.e., 00010. It is the negative number and the final answer.